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| **Fiscal Year:** | FY 2017 | **Task Last Updated:** | FY 02/27/2017 |
| **PI Name:** | Simpson, Richard Ph.D. | | |
| **Project Title:** | The Impact of Modeled Microgravity and Prior Radiation Exposure on Cytomegalovirus Reactivation and Host Immune Evasion | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Space Human Factors Engineering | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | (1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 85721-0001 | **Congressional District:** | 3 |
| **Comments:** | NOTE: Formerly at University of Houston until September 2017 move to University of Arizona. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI) |
| **Start Date:** | 10/21/2015 | **End Date:** | 10/20/2018 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 5 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Norsk, Peter | **Contact Phone:** | |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 10/20/2018 per NSSC information (Ed., 3/5/18)<br>NOTE: Element change to Human Health Countermeasures; previously Space Human Factors & Habitability (Ed., 1/18/17) | | |
| **Key Personnel Changes/Previous PI:** | N/A | | |
| **COI Name (Institution):** | Crucian, Brian Ph.D. ( NASA Johnson Space Center )<br>Laughlin, Mitzi Ph.D. ( University of Houston )<br>Markofski, Melissa Ph.D. ( University of Houston )<br>Mehta, Satish Ph.D. ( NASA Johnson Space Center )<br>Pierson, Duane Ph.D. ( NASA Johnson Space Center )<br>Rezvani, Katayoun M.D., Ph.D. ( The University of Texas MD Anderson Cancer Cente ) | | |
| **Grant/Contract No.:** | NNX16AB29G | | |
| **Performance Goal No.:** | | | |

**Performance Goal Text:**

**Task Description:**

Cytomegalovirus (CMV), which infects 50-80% of the US population, and CMV DNA is frequently found in the body fluids of astronauts during space travel indicating that the spaceflight environment can initiate active CMV infections. The effects of microgravity on CMV replication and its potential to evade the host immune system are unknown. In this study, we will utilize the rotating wall vessel (RWV) cell culture system to simulate microgravity on Earth allowing us to study the virulent and infectivity properties of CMV and how it interacts with host immune cells in the absence of gravity. The study hypothesizes that modeled microgravity will evoke changes in cellular gene expression that will lead to CMV reactivation and increase virus replication and infectivity. The researchers believe that modeled microgravity will impair the anti-CMV responses of host T-cells and NK-cells that will allow viral persistence and replication to occur. During testing both traditional and cutting-edge laboratory techniques that are typically used to assess virulence and immune cell function in the transplant patient setting allowing determination if CMV infection in the microgravity environment is likely to increase the risk of an adverse health event during exploration class spaceflight missions.
Specific Aims:

To determine the impact of simulated microgravity on CMV replication and its potential to evade the host immune system

a) To determine the effects of modeled microgravity on CMV replication and the expression of genes associated with latency and lytic activity.

b) To determine the impact of modeled microgravity on the anti-CMV activity of host T-cells and NK-cells

This study will provide preliminary data to determine if CMV virus displays increased infectivity when exposed to a low shear modeled microgravity environment. Data will be evaluated and added to the evidence base capturing the responses to microorganisms grown in rotating wall vessels.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Our early results indicate that simulated microgravity might actually improve the ability of T-cells to expand when stimulated with cytomegalovirus peptides. This could have important implications for immunotherapy that relies on methods to increase the number of T-cells reacting to specific virus and tumor antigens so that they may be given to immunocompromised patients after stem cell and solid organ transplantation.

**Task Progress:**

Year 1 objectives for this project were:
1. Determine the effects of simulated microgravity on the function of NK-cells and Virus-Specific T-cells ; 2. Establish and optimize the in vitro infection model using live CMV

Objective #1: Determine the effects of simulated microgravity (SMG) on the function of NK-cells and Virus-Specific T-cells

Summary: Specific Aim 2 of this project was to determine the impact of modeled microgravity and radiation on the anti-CMV activity of host T-cells and NK-cells. Our hypothesis was that, compared to traditional cell culture, simulated microgravity (SMG) would impair CMV-specific T-cell and NK-cell expansion and reduce the effectiveness of these cells in curtailing CMV replication and infectivity. In year 1, we performed a number of time series experiments to determine the maximum length of time primary human blood NK-cells can be cultured in the rotating wall vessels without significant reductions in cell viability (<95%). Our goal was to use primary human NK-cells without any form of ex vivo manipulation (i.e., survival cytokines such as IL-15 and IL-2) as this would introduce another variable that could be impacted by SMG. Our initial experiments determined that 12 h was the optimal length of time for NK-cells to spend in SMG without causing reductions in cell viability. This length of time exposed to SMG was also found to cause changes in NK-cell function that was not observed when the NK-cells were cultured for the same length of time in static or horizontal rotation control conditions.

Our early conclusions from this work are as follows:

1. Short-term exposure to simulated microgravity impairs NK-cell function. This was confirmed using two different analytical methods and assays.

2. The ability of NK-cells to form conjugates (physical connections with their target cells) is not impaired with SMG and is therefore unlikely to be the mechanism by which NK-cell function is inhibited by SMG.

3. SMG appear to reduce the ability of NK-cells to secrete pro-inflammatory cytokines that are required to stimulate NK-cell mediated killing and also their expression of cytolytic granules that are important in the process of NK-cell killing. It is possible, therefore, that SMG is ‘disarming’ NK-cells and, reducing their killing potential.

Our current objectives are to:

1. Determine if SMG impairs the ability of NK-cells to kill CMV-infected autologous target cells. We will achieve this aim initially by using autologous PHA blast cells pulsed with CMV viral peptides spanning the entire pp65 and IE-1 CMV antigens before proceeding to autologous target cells infected with live CMV ; 2. Determine the effects of SMG on the ability of NK-cells to control CMV infection in vitro.

The Effects of Simulated Microgravity on CMV-specific T-cell Responses

Peripheral blood mononuclear cells obtained from CMV-infected donors were placed in bioreactor vessels under three different conditions: (i) static control; (ii) horizontal rotation control; or (iii) vertical rotation/simulated microgravity. The horizontal control condition was used to account for potential effects of fluid shear stress that are independent of SMG. The bioreactors were incubated for 12 h in a cell culture cabinet at 37°C and 5% CO2. Following incubation, the PBMCs were pulsed with viral peptides spanning the entire CMV pp65 and IE-s antigens and expanded in G-REX bioreactors for 14-days. The frequency of CMV-specific T-cells in the PBMCs before and after expansion was determined using a 12 h IFN-gamma ELISPOT assay. Contrary to our hypothesis, our early findings indicate that SMG actually enhances the ex vivo expansion of CMV-specific T-cells. Indeed, the number of T-cells reacting with the CMV

pp65 and IE-1 antigens is greater after SMG compared to both static and horizontal conditions. These experiments have only been replicated a few times (n=4) so the enhancing effects of SMG on CMV-specific T-cell expansion are not currently significant (pp65: p=0.09; IE-1: p=0.22) so we will attempt to replicate these findings on several more participants.

Our early conclusions from this work are as follows:

1. Short-term exposure to simulated microgravity may enhance the ex vivo expansion of T-cells recognizing the CMV antigens pp65 and IE-1. Further experiments will confirm or refute these initial observations.

Our current objectives are to:

1. Determine if SMG affects the ability of CMV-specific T-cells to recognize and kill autologous target cells pulsed with CMV-peptides and infected with live CMV ; 2. Determine the T-cell subtypes (and their phenotypes) affected by SMG with regards to expansion and cytotoxic function. This will be examined using both flow cytometry and Cy-TOF.

Objective #2: Establish and optimize the in vitro infection model using live CMV

Summary: Specific Aim 1 of this project is to determine the effects of modeled microgravity on CMV replication and the expression of genes associated with latency and lytic activity. Our hypothesis was that, compared to traditional cell culture, simulated microgravity will increase lytic gene expression in CMV-infected monocytes and amplify viral replication and infectivity upon monocyte differentiation. Before moving our experiments to primary human monocytes, we deemed it important to first of all establish the CMV infection model using a suitable cell line. For this work we have been using the Kasumi-3 cell line, which has been shown to be a tractable in vitro model system to study CMV latency and reactivation. We initially determined if the Kasumi-3 cells would be suitable for simulated microgravity by performing a time-series survival experiment. The Kasumi-3 cells remained viable in SMG for up to 16-days and were greater/comparable to static control conditions. Both TOWNE and FIX strains of CMV were propagated in MRC-5 cells and used to infect Kasumi-3 cells. We recently confirmed that our methods were successful to infect Kasumi-3 cells with CMV and are currently confirming the time course required for the virus to establish latency. When this has been attained, we will attempt to reactivate CMV from the infected Kasumi-3 cells by stimulating them with PMA and quantifying viral load in the cell supernatants as part of a time series experiment. When this has been established we will proceed to the SMG experiments.

Our current objectives are to:

1. Determine if SMG affects the ability of CMV-infected Kasumi-3 cells to reactivate CMV when stimulated with PMA ; 2. Determine if SMG alters expression of genes associated with CMV latency and reactivation.

| Bibliography Type: | Description: (Last Updated: 01/12/2026) |
|---|---|
| Articles in Peer-reviewed Journals | Simpson RJ, Bigley AB, Spielmann G, LaVoy EC, Kunz H, Bollard CM. "Human cytomegalovirus infection and the immune response to exercise." Exerc Immunol Rev. 2016;22:8-27. PubMed PMID: 26853134 , Feb-2016 |
| Articles in Peer-reviewed Journals | Bigley AB, Rezvani K, Shah N, Sekine T, Balneger N, Pistillo M, Agha N, Kunz H, O'Connor DP, Bollard CM, Simpson RJ. "Latent CMV infection enhances anti-tumor cytotoxicity through accumulation of NKG2C+ NK-cells in healthy humans." Clin Exp Immunol. 2016 Aug;185(2):239-51. http://dx.doi.org/10.1111/cei.12785 ; PubMed PMID: 26940026; PubMed Central PMCID: PMC4955006 , Aug-2016 |
| Articles in Peer-reviewed Journals | Spielmann G, Bollard CM, Kunz H, Hanley PJ, Simpson RJ. "A single exercise bout enhances the manufacture of viral-specific T-cells from healthy donors: implications for allogeneic adoptive transfer immunotherapy." Sci Rep. 2016 May 16;6:25852. http://dx.doi.org/10.1038/srep25852 ; PubMed PMID: 27181409; PubMed Central PMCID: PMC4867645 , May-2016 |
| Articles in Peer-reviewed Journals | LaVoy EC, Hussain M, Reed J, Kunz H, Pistillo M, Bigley AB, Simpson RJ. "T-cell redeployment and intracellular cytokine expression following exercise: effects of exercise intensity and cytomegalovirus infection." Physiol Rep. 2017 Jan;5(1):e13070. Epub 2017 Jan 13. http://dx.doi.org/10.14814/phy2.13070 ; PubMed PMID: 28087817; PubMed Central PMCID: PMC5256156 , Jan-2017 |